Project Design Phase-I
Problem-Solution Fit Template

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Team ID	698c67ecb31ff1dfca225748
Project Name	PlantCare AI: Advanced Plant Disease Detection Using Transfer Learning
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Maximum Marks	4 Marks

Problem-Solution Fit Canvas

The Problem-Solution Fit Canvas maps the real-world problem of plant disease diagnosis against PlantCare AI's solution, validating product-market fit before full development investment.

3 — TRIGGERS (TR)

• Farmer sees their neighbour's crop failing rapidly from an unidentified disease — triggers panic and urgency for fast diagnosis
• News reports of large-scale crop disease outbreaks in the region (e.g., wheat rust, tomato blight season) — triggers proactive monitoring behaviour
• Farmer notices unusual yellowing, spots, or wilting on leaves — triggers manual inspection followed by desire for expert confirmation
• Agricultural extension officer visit reveals diseased crops on neighbouring farms — triggers awareness and urgency in nearby farmers
• Post-harvest loss due to previously undiagnosed disease — triggers desire for early detection tools to prevent future losses
• Social media / WhatsApp groups sharing photos of diseased crops asking for identification — triggers community-based diagnosis seeking behaviour
• Government agricultural advisory warning about seasonal disease outbreak risk — triggers proactive checking of own crops

10 — YOUR SOLUTION (SL)

PlantCare AI is a web-based AI plant disease detection application that classifies leaf photographs into 38 plant health categories (26 diseases + 12 healthy states) within 1.5 seconds using a fine-tuned MobileNetV2 deep learning model trained on 87,000+ images.

How it works (3 steps):
Step 1 — UPLOAD: User photographs a diseased leaf with any smartphone or camera and uploads it via the PlantCare web interface (drag-and-drop or file picker).

Step 2 — AI ANALYSIS: MobileNetV2 model preprocesses the image (resize to 224×224, MobileNetV2 normalization), runs inference, and maps the argmax prediction to a disease class label via class_names.json.

Step 3 — GET RESULTS: result.html renders the uploaded image, a RED badge ('DISEASE DETECTED') or GREEN badge ('HEALTHY PLANT'), the plant name, disease name, and specific recommended treatment/care actions.

Key solution parameters: Python + Flask backend | MobileNetV2 (3.4M params, 96.6% accuracy) | No GPU required for inference | Free to use | Deployable anywhere with Python installed.

8 — CHANNELS OF BEHAVIOUR (CH)

8.1 ONLINE CHANNELS:

• Web Browser Access: Primary channel — users access PlantCare AI via any browser on desktop or mobile at http://[domain].com. No app installation required.

• WhatsApp / Telegram Sharing: Farmers share disease diagnosis results (screenshot of result page) in agriculture community groups for peer validation and collective action.

• YouTube Agricultural Channels: Tutorial videos demonstrating PlantCare AI usage reach farmer YouTube audiences in regional languages.

• Agricultural Extension Service Portals: Integration with government e-Agriculture portals (e.g., India's eNAM, Kenya's DigiFarm) as embedded diagnostic tool.

• Social Media (Facebook Groups): Farming community Facebook groups — farmers share results and tag affected neighbours.

8.2 OFFLINE CHANNELS:

• Agricultural Extension Officer Visits: Extension officers demonstrate PlantCare AI on tablets during field visits — act as offline channel through which farmers adopt the tool.

• Farmer Training Workshops: Agri-NGOs and cooperatives conduct workshops demonstrating the application; printed QR codes link to the web app.

• Farm Service Centres / Input Dealers: Pesticide retailers and seed dealers show PlantCare AI results on shop computers to recommend targeted treatments (instead of broad-spectrum chemicals).

• School / College Agricultural Programs: Agricultural colleges include PlantCare AI in practical curriculum — students share with farming families.

4 — EMOTIONS: BEFORE / AFTER (EM)

EMOTIONS BEFORE (When farmer faces plant disease problem):

• ANXIOUS: Unsure whether the spotted/yellowing leaves indicate a serious disease or normal environmental stress — paralysed by uncertainty

• HELPLESS: No agronomist available locally; nearest agriculture office may be hours away; can't afford a private consultant

• FINANCIALLY STRESSED: Fear of total crop loss and associated income loss — especially for smallholder farmers with no crop insurance

• OVERWHELMED: Multiple plants showing different symptoms; farmer doesn't know which to treat first or how urgently

• DISTRUST: Past experience of misdiagnosis from non-expert advice led to wrong pesticide purchase and wasted money

EMOTIONS AFTER (After using PlantCare AI):
• CONFIDENT: Clear disease identification gives farmer a specific name to research, discuss with others, and act upon decisively
• EMPOWERED: Access to expert-level diagnosis at zero cost — democratized AI puts the farmer in control of their own crop health
• RELIEVED: Immediate result (under 2 seconds) eliminates days of anxious uncertainty; actionable recommendations remove decision paralysis
• COST-CONSCIOUS SATISFACTION: Targeted treatment recommendation prevents buying wrong or excess chemicals — tangible financial saving
• CONNECTED: Result page and sharing capability connect farmer to community knowledge — feeling of not being alone in facing the problem
• INFORMED: Understanding of disease name, mechanism, and recommended actions builds farmer's long-term agricultural knowledge

Solution Fit Validation Summary

Canvas Element	PlantCare AI Fit Statement
Problem-Solution Fit	The trigger (farmer sees diseased crop) maps directly to the solution action (photograph + upload). The 1.5-second result time matches the urgency of the problem.
Channel Fit	WhatsApp community sharing aligns with farmer social behaviour. Browser-based app (no install) fits low-tech-literacy rural users on basic smartphones.
Emotion Fit	Confident + Empowered post-use emotions are the inverse of Anxious + Helpless pre-use emotions — strong emotional transformation validates the solution value proposition.
Scalability Fit	Solution starts as a web app, scales to API, mobile app, and IoT integration — matching all identified offline and online channels progressively.
Social Impact Fit	Free access model ensures the highest-need users (subsistence farmers) are not excluded by cost — aligns with UN SDG 2 (Zero Hunger) and SDG 15 (Life on Land).